PROJECT-BASED LEARNING

PHASE-I EVALUATION

CAMPUSXPLORE

Campus Navigation and Building Information System

Team ID: DSCPP-III-2026-T039

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T039
Semester: 3rd
Section: D,B,G,DS1
Domain: DSA and OOP using C++

Team Members

1. Garvit Aggarwal – 2510011570
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Mentor

Dr.Pawan Kumar Mishra

Interactions completed: 3

Problem Statement & Motivation

Problem Statement

Campus users can struggle to find rooms and facilities indoors. GPS is unreliable inside buildings, while some solutions need extra infrastructure. CampusXplore uses manual or QR-based location input with graph-based shortest-path routing without BLE.

Motivation

- GPS is unreliable indoors.
- New users may struggle to locate destinations.
- Some methods need extra infrastructure.

Target Users / Stakeholders

- Students, new users, and campus visitors.
- Faculty and campus administration.
- Users needing accessible or emergency routes.

Objectives

Project Objectives

1. Identify the starting point manually or through a supported QR code.
2. Model campus locations as a graph and compute shortest paths using Dijkstra's algorithm and a priority queue.
3. Support multi-floor movement, accessibility preferences, blocked routes, and emergency destinations.
4. Reconstruct and display the route on a simple 2D or 2.5D map.
5. Demonstrate data-structure and OOP concepts using C++.

Key Objective: Provide an efficient shortest route from a selected starting point to a destination.

Proposed Solution

Proposed Approach

1. **Input:** Manual entry, list selection, or QR scan.
2. **Graph:** Adjacency list for campus locations.
3. **Routing:** Dijkstra's algorithm with a min-heap priority queue.
4. **Dynamic routing:** Ignore blocked edges and recalculate.
5. **Output:** Reconstruct and display the route on a 2D/2.5D map.

Key Features

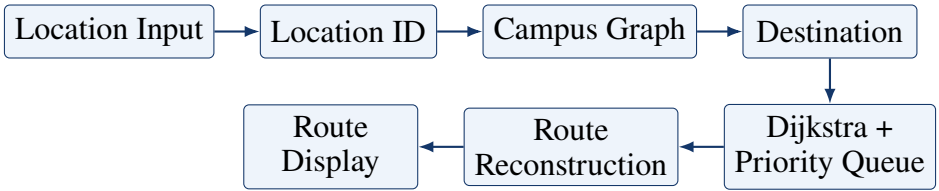
- QR-based or manual location identification.
- Destination search and shortest-path navigation.
- Multi-floor routing using stairs and elevators.
- Accessibility-aware route choices.
- Temporary blocks and emergency rerouting.

Integration of PBL Course Concepts

Course	Concepts Applied	Project Use
Data Structures in C	Graphs, adjacency list, priority queue	Graph model and routing
OOPs with C++	Classes, objects, STL structures	Navigation system design
Operating Systems	Not specified in the proposal –	–
DBMS	Not specified in the proposal	–

Course integration: Graph-based DSA and C++ OOP concepts are directly applied to the navigation prototype.

System Architecture / Workflow



Major Components

- Location input and destination selection.
- Campus graph: rooms, entrances, stairs and elevators.
- Dijkstra routing, route reconstruction and display.

Data / Control Flow

- Input becomes valid start and destination IDs.
- Routing applies accessibility and blockage constraints.
- The path is reconstructed and displayed to the user.

Technology Stack & Methodology

Technology Stack

- Programming: C++
- Core structure: Graph / adjacency list
- Algorithm: Dijkstra + min-heap priority queue
- UI: Simple 2D / 2.5D map
- Other tools: VS Code, GitHub

Development Methodology

1. Requirement analysis
2. System and graph design
3. Implementation
4. Testing
5. Evaluation and demo

Initial Progress & Team Contribution

Progress Achieved

- Literature study completed.
- Requirements and navigation features identified.
- High-level architecture and workflow prepared.
- Campus graph and location-data structure defined.
- Initial implementation status not specified in the proposal.

Individual Contribution

- **Garvit Aggarwal** – Project Planning and Team Coordination.
- **Raj Maletha** – Testing and Debugging.
- **Ranveer Singh Bisht** – OOP and C++ Implementation.
- **Arnav Chamoli** – Campus Mapping and Graph Design.

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Proposal & Design
2. Phase-II: Development
3. Phase-III: Final Implementation

Expected Outcomes

1. Working C++ prototype.
2. Shortest-path routing.
3. Multi-floor + accessibility.
4. Dynamic rerouting + emergencies.

References

1. L. E. Gryzun, O. V. Shcherbakov, and B. O. Bida, "Development of the information system for navigation in modern university campus," CTE Workshop Proceedings, vol. 10, pp. 380–398, 2023, doi: 10.55056/cte.566.
2. P. Switalski and A. Salamończyk, "Universal structural map for indoor navigation in university campus," Studia Informatica. Systems and Information Technology, no. 2 (29), 2023, doi: 10.34739/si.2023.29.05.
3. O. A. Gbadamosi, A. O. Ilori, M. A. Oduwale, and O. O. Abiona, "Development of a smart GIS-based campus navigation and shortest path routing system for OAUSTECH, Ondo State, Nigeria," Review of Information Engineering and Applications, vol. 12, no. 1, pp. 48–58, 2025, doi: 10.18488/79.v12i1.4656.